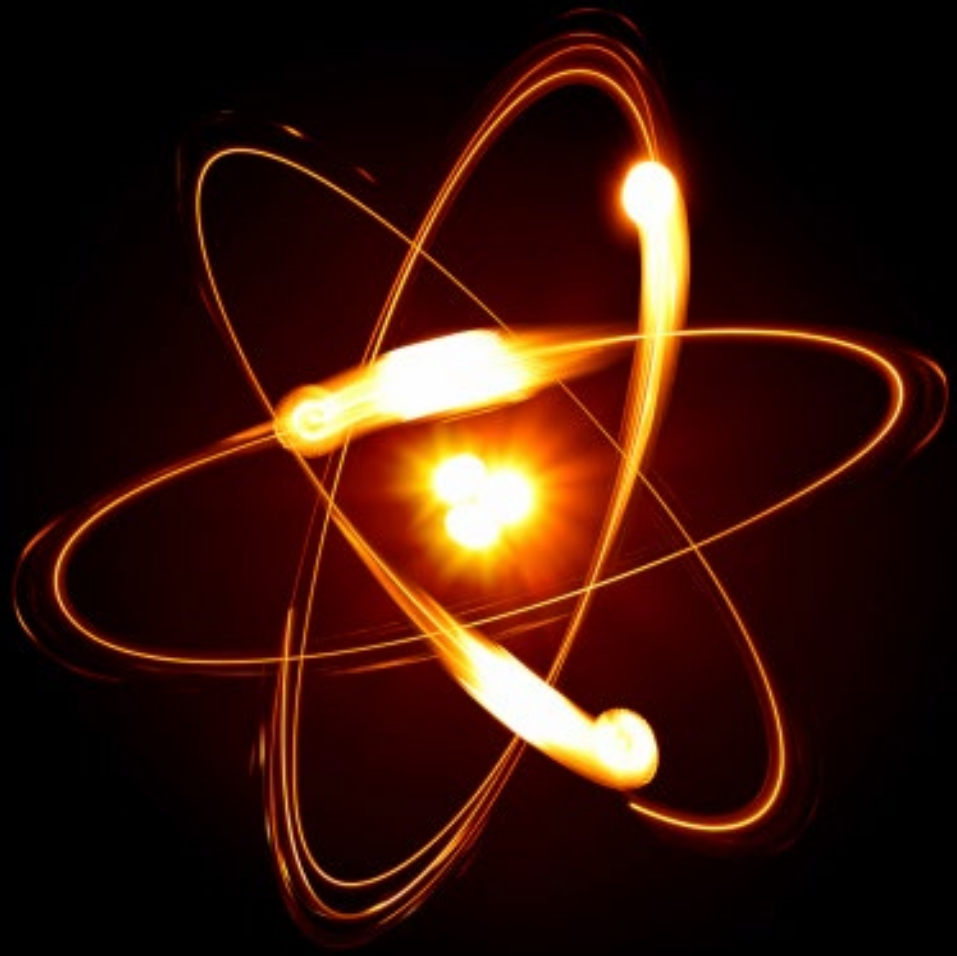
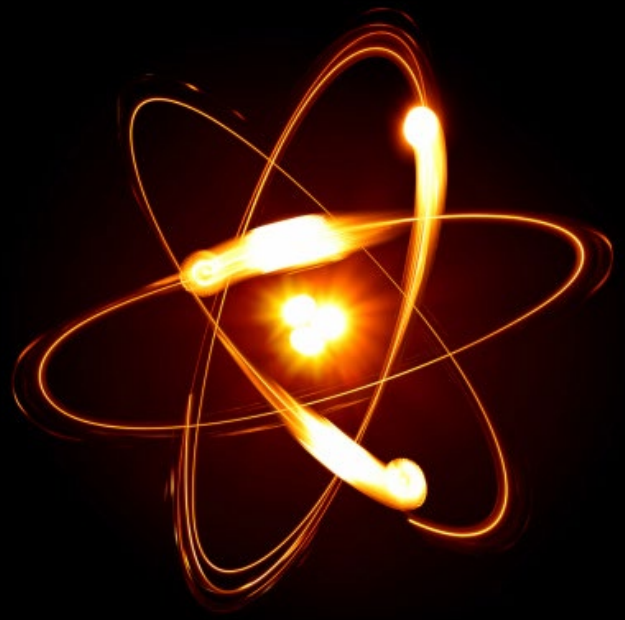
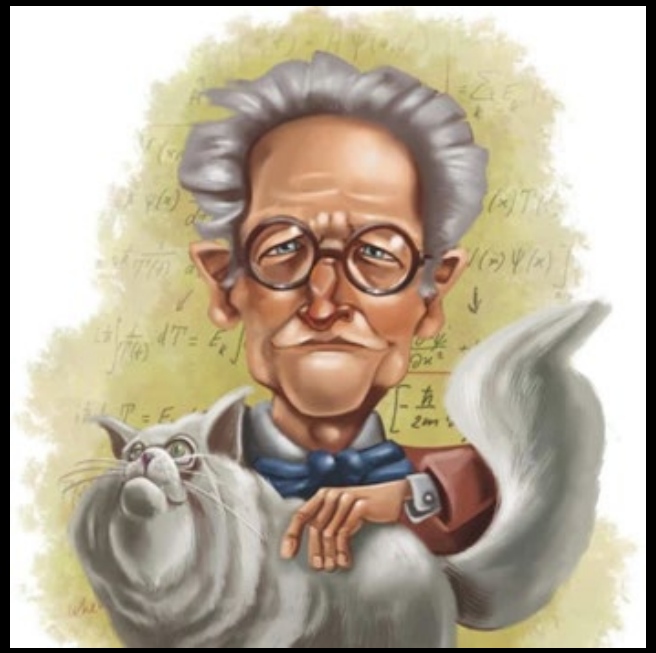
Chapter 6.

Schrödinger's Theory of Quantum Physics



Beatriz González del Río





Chapter 6.

Schrödinger's theory of quantum physics

- Introduction
- Wave equations in classical physics.
- Schrödinger's equation
- Born's interpretation of wave functions
- Expectation values, momentum space, uncertainties
- Time-independent Schrödinger's equation.
- Stationary states and energy quantization

Introduction

- Microscopic particles behave as if its motion was ruled by a wave (interference, diffraction,)
- In some simple situations (free particles), the wave has been found by using de Broglie's postulate.
- The interesting systems are those in which the particle(s) is(are) under the action of some potential.

What should we do?

- Find a general approach to calculate the wave associated to the particle: Schrödinger's wave equation
- This approach must provide some connection between the particle-like and wave-like aspects.
- The de Broglie postulate must be a particular case.
- Newton's equations must be the macroscopic limit.

Wave equations in classical physics

Sound propagation in a fluid

Homogeneous fluid in equilibrium and be P_0 its pressure. At t=0 there is some perturbation, $p'(\vec{r},t)$, which propagates, as a sound wave, in the fluid. Then at time t in point $\vec{r}$, the pressure will be $P_0 + p'(\vec{r},t)$. The perturbation propagates according to the equation (v is the sound velocity in the fluid)

$$\Delta p'(\vec{r},t) = \frac{1}{v^2}\frac{\partial^2}{\partial t^2} p'(\vec{r},t)$$

Electromagnetic waves

Electromagnetic radiation in the vacuum. Then at time t in point $\vec{r}$ there are electric $\vec{E}(\vec{r}\,,\,t) = (E_x, E_y, E_z)$ and magnetic $\vec{B}\,(\vec{r}\,,\,t) = (B_x, B_y, B_z)$ fields that propagate according to

$$\Delta E_x(\vec{r},t) = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} E_x(\vec{r},t)$$

Same for $E_y(\vec{r},t)$ and $E_z(\vec{r},t)$.

Wave equations in classical physics

Main properties of the wave equations

- For the classical wave equations considered here, the velocity of propagation is independent of wavelength, λ, and frequency ν ($\lambda\nu = v$). Wave front moving in the direction given by the unit vector $\vec{u}$.

$$p'(\vec{r},t) \equiv p'(\sigma) \qquad \sigma = \vec{r}\cdot\frac{\vec{u}}{\lambda} - \nu t$$

For example, take $p'(\sigma) = a\cos\sigma$

$$\Delta p'(\vec{r},t) = \frac{d^2p'}{d\sigma^2}\frac{1}{\lambda^2} \qquad \frac{\partial^2}{\partial t^2}p'(\vec{r},t) = \frac{d^2p'}{d\sigma^2}(-\nu)^2$$

and the wave equation is satisfied

- **Superposition principle**. If $p'_1(\vec{r},t)$ and $p'_2(\vec{r},t)$ are two sound perturbations, then $p'(\vec{r},t) = C_1p'_1(\vec{r},t) + C_2p'_2(\vec{r},t)$ is also a solution of the wave equation and represents another sound perturbation.

  The same applies for electromagnetic waves.

- $p'(\vec{r},t)$ and $E_x(\vec{r},t)$ are physical magnitudes described by **real** functions.

$$p'(\vec{r},t) = a\cos 2\pi\left(\vec{r}\cdot\frac{\vec{u}}{\lambda} - \nu t\right) = a\cos\left(\vec{r}\cdot\vec{k} - \omega t\right)$$

Schrödinger´s equation

$$-\frac{\hbar^2}{2m}\Delta\Psi(\vec{r},t) = i\hbar\frac{\partial\Psi(\vec{r},t)}{\partial t}$$



In general: (particle with potential energy $V(\vec{r},t)$)

$$-\frac{\hbar^2}{2m}\Delta\Psi(\vec{r},t) + V(\vec{r},t)\Psi(\vec{r},t) = i\hbar\frac{\partial\Psi(\vec{r},t)}{\partial t}$$

In One-dimension

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t) = i\hbar\frac{\partial\Psi}{\partial t}$$

Born's interpretation of wave functions

Assume $\Psi(\vec{r},t)$ is the wave function of a particle. Does it have any physical meaning?

$$\Psi(\vec{r},t) = \mathfrak{Re}\,\Psi(\vec{r},t) + i\,\mathfrak{Im}\,\Psi(\vec{r},t)$$

Reasonable conditions on $\Psi(\vec{r},t)$

- $\Psi(\vec{r},t)$ should take significant values for $\vec{r}$ around the particle's position.
- $\Psi(\vec{r},t)$ should vanish for $\vec{r}$ far from the particle.

How to measure the magnitude of $\Psi(\vec{r},t)$? There are several choices, i.e.

$$|\Psi(\vec{r},t)|^2 = |\mathfrak{Re}\,\Psi(\vec{r},t)|^2 + |\mathfrak{Im}\,\Psi(\vec{r},t)\,|^2$$ at a point $\vec{r}$ and t

$$\int_{\mathcal{R}} |\Psi(\vec{r},t)|^2 d\vec{r}$$ in a region $\mathcal{R}$ and t

Therefore, we impose:

$$\int_{\mathcal{R}} |\Psi(\vec{r},t)|^2 d\vec{r} = N \leq \infty$$

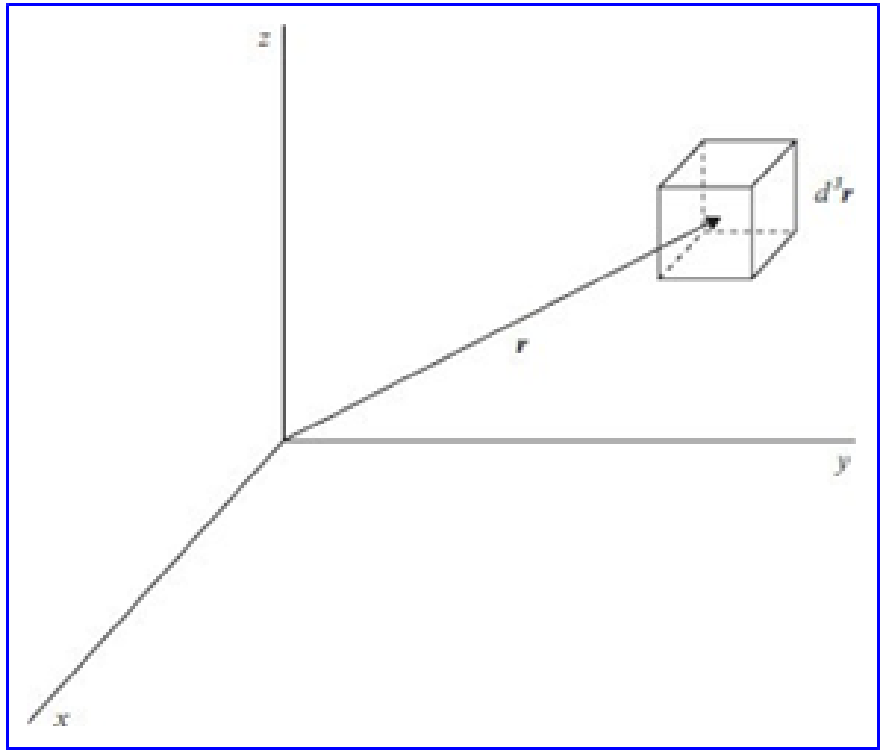
Born's interpretation of wave functions

M. Born (1926) suggests the following **physical meaning**:

Let $d\vec{r}$ be a small volume element around point $\vec{r}$ and assume that $\int_{all\ space} |\Psi(\vec{r},t)|^2 d\vec{r} = N$

$\frac{1}{N}|\Psi(\vec{r},t)|^2 d\vec{r} \Rightarrow$ Probability of finding the particle at time *t* within the small volume $d\vec{r}$, centered around $\vec{r}$.

$\frac{1}{N}\int_{\mathcal{R}} |\Psi(\vec{r},t)|^2 d\vec{r} \Rightarrow$ Probability of finding the particle at time *t* within the region $\mathcal{R}$.



Now, the matter wave is something like a **probability wave**

Now, $|\Psi(\vec{r},t)|^2$ plays the role of a probability density function (p.d.f) associated to the continuous random variable $\mathcal{R}$

However, a plane wave can't be normalized and therefore the previous probabilistic interpretation can't be applied.

Born's interpretation of wave functions

Schrödinger's Equation + Probabilistic interpretation ⇒ Basis of Quantum Mechanics



Newton's Laws ⇒ Basis of Classical Mechanics

According to the standard interpretation of QM the **wave function $\Psi(\vec{r}, t)$ contains ALL the information that can be obtained from the particle.**

Therefore, it is said that $\Psi(\vec{r}, t)$ represents **the state of the particle.**

Review of classical mechanics (CM)

Consider a particle of mass *m* moving in a potential $V(\vec{r},t)$. At any time, *t*, its position and linear momentum are given by the vectors $\vec{r} = \big(x(t), y(t), z(t)\big)$ and $\vec{p}(t) = \Big(p_x(t), p_y(t), p_z(t)\Big)$, respectively.

The total energy of the particle, kinetic plus potential, is given by

$$\mathcal{H}(\vec{r},\vec{p},t) = \frac{p^2}{2m} + V(\vec{r},t)$$

where $\mathcal{H}(\vec{r},\vec{p},t)$ is the classical hamiltonian.

The time evolution of the particle's state is completely specified by the Hamilton equations.

$$\frac{d\vec{r}}{dt} = \nabla_{\vec{p}}\mathcal{H}(\vec{r},\vec{p},t) \qquad \frac{d\vec{p}}{dt} = -\nabla_{\vec{r}}\mathcal{H}(\vec{r},\vec{p},t\ )$$

They are first order differential eqns. whose solution $(\vec{r}(t),\vec{p}(t))$ is unique, provided we know its values at a given, initial time, e.g. $(\vec{r}(t_0),\vec{p}(t_0))$.

In other words, the state of the system at any time is completely specified if we know its state at an initial time.

Review of classical mechanics (CM)

In terms of the components, the Hamilton equations are

$$\frac{dx}{dt} = \frac{\partial \mathcal{H}(\vec{r},\vec{p},t)}{\partial p_x} = \frac{p_x}{m} \qquad \frac{dp_x}{dt} = -\frac{\partial \mathcal{H}(\vec{r},\vec{p},t)}{\partial x} = -\frac{\partial V(\vec{r},t)}{\partial x}$$

and similarly for *dy/dt, dz/dt, dp$_y$/dt, dp$_z$/dt*

From the previous eqns, we obtain Newton's eqn.

$$m\frac{d^2\vec{r}}{dt^2} = -\nabla_{\vec{r}} V(\vec{r},t)$$

On the other hand, it can be shown that $\frac{d\mathcal{H}(\vec{r}(t),\vec{p}(t),t)}{dt} = \frac{\partial V(\vec{r},t)}{\partial t}$

therefore, if V does not explicitly depend on time, that is $V(\vec{r},t) \equiv V(\vec{r})$, then

$$\frac{d\mathcal{H}(\vec{r}(t),\vec{p}(t),t)}{dt} = 0 \quad \Rightarrow \quad \mathcal{H}(\vec{r}(t),\vec{p}(t),t) = constant$$

the total energy remains constant.

Review of classical mechanics (CM)

The classical description of a particle proceeds as

- The state of the system at t_0 is characterized by the $\vec{r}(t_0)$ and $\vec{p}(t_0)$.

- The value at t_0 of ALL the physical magnitudes is COMPLETELY determined when the state is known.

- The time evolution of the system's state is completely specified by the Hamilton equations. They are first order differential eqns. whose solution ($\vec{r}(t)$ and $\vec{p}(t)$) is unique once we know its values at a given time, e.g. ($\vec{r}(t_0)$ and $\vec{p}(t_0)$). In other words, the state of the system at any time is completely specified if we know its state at an initial time.

Expectation values

What about quantum physics? How to gather information about different magnitudes?

Remember that according to the standard interpretation of QM the wave function $\Psi(\vec{r},t)$ contains **ALL** the information that can be obtained from the particle.

How can we extract it?

1-D. Particle characterized by $\Psi(x,t)$

Consider a **normalized** $\Psi(x,t)$. The particle may be at different positions *x* with respective probabilities $|\Psi(x,t)|^2$. This is the equivalent of a probability density function (p.d.f.) associated to a continuous random variable, X. Notice that, in principle, X may take any value within the range $(-\infty, \infty)$

Now, if we compute the mean value

$$< x >= \int dx\, x\, |\Psi(x,t)|^2 \equiv \int dx\, \Psi^*(x,t)\, x\, \Psi(x,t)$$

If we now consider any function, F(x), of the random variable

$$< F(x) >= \int dx\, \Psi^*(x,t)\, F(x)\, \Psi(x,t)$$

Not normalized? Work it out !!

Expectation values

3-D. Consider a normalized $\Psi(\vec{r},t)$. The particle may be at different positions $\vec{r}$ with respective probabilities $|\Psi(\vec{r},t)|^2$. This is the equivalent of a probability density function. Now,

$$< \vec{r} >= \int d\vec{r}\, \Psi^*(\vec{r},t)\, \vec{r}\, \Psi(\vec{r},t) \equiv \int d\vec{r}\, \vec{r} |\Psi(\vec{r},t)|^2 \equiv (< x >, < y >, < z >)$$

$$< x >= \int d\vec{r}\, \Psi^*(\vec{r},t)\, x\, \Psi(\vec{r},t) \qquad < y >= \int d\vec{r}\, \Psi^*(\vec{r},t)\, y\, \Psi(\vec{r},t) \qquad < z >= \int d\vec{r}\, \Psi^*(\vec{r},t)\, z\, \Psi(\vec{r},t)$$

If $< x >$, does that mean that a measurement of position must give $x = 0$?

For any function of $\vec{r}$, e.g. $F(\vec{r})$, we have

$$< F(\vec{r}) >= \int d\vec{r}\, \Psi^*(\vec{r},t)\, F(\vec{r})\, \Psi(\vec{r},t)$$

Expectation values

Similar arguments can be applied for the linear momentum. (**1-D**)

There will also be a mean value $< p_x >$

$$< p_x >= \int dx\, \Psi^*(x,t)\, p_x\, \Psi(x,t)$$??

There must be some connection between the classical laws of motion for a macroscopic particle and the equations describing the time evolution of the mean values for a microscopic particle.

Time evolution of the position's mean value

$$\frac{d}{dt} < x >= \frac{1}{m} \int dx\, \Psi^*(x,t) \left(-i\hbar \frac{d}{dx}\right) \Psi(x,t)$$

Comparison with the classical result, $dx/dt = p_x/m$, allows to identify

$$\boxed{< p_x >= \int dx\, \Psi^*(x,t) \left(-i\hbar \frac{d}{dx}\right) \Psi(x,t)}$$

Not normalized? Work it out !!

What about $< p_x^2 >$?

Expectation values

Similar arguments can be applied in 3-D

Now, the mean value is $< \vec{p} > \equiv (< p_x >, < p_y >, < p_z >)$

Recall the time evolution of the position's mean value (now in 3-D)

$$\frac{d}{dt} < x > = \frac{1}{m} \int d\vec{r}\, \Psi^*(\vec{r},t) \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi(\vec{r},t)$$

and by comparison with the classical result $dx/dt = p_x/m$ leads to

$$< p_x > = \int d\vec{r}\, \Psi^*(\vec{r},t) \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi(\vec{r},t)$$

and similarly

$$< p_y > = \int d\vec{r}\, \Psi^*(\vec{r},t) \left(-i\hbar \frac{\partial}{\partial y} \right) \Psi(\vec{r},t) \qquad < p_z > = \int d\vec{r}\, \Psi^*(\vec{r},t) \left(-i\hbar \frac{\partial}{\partial z} \right) \Psi(\vec{r},t)$$

In short $< \vec{p} > = \int d\vec{r}\, \Psi^*(\vec{r},t)(-i\hbar\nabla)\Psi(\vec{r},t)$

Expectation values

Let's consider the time evolution of the momentum's mean value (**1-D**)

$$\frac{d}{dt} < p_x >= \frac{d}{dt} \int dx\, \Psi^*(x,t) \left(-i\hbar \frac{\partial}{\partial x}\right) \Psi(x,t) = \int dx\, \Psi^*(x,t) \left(-\frac{\partial V(x,t)}{\partial x}\right) \Psi(x,t)$$

which allows to identify $\frac{d}{dt} < p_x >=< -\frac{\partial V(x,t)}{\partial x} >$

Now in **3-D** we generalize: $\frac{d}{dt} < p_x >=< -\frac{\partial V(\vec{r},t)}{\partial x} >$

with similar relations holding for $\frac{d}{dt} < p_y >$ and $\frac{d}{dt} < p_z >$.

Therefore, we summarize the previous equations as

$$\boxed{\frac{d}{dt} < \vec{p} >=< -\nabla V(\vec{r},t) >}$$

Expectation values

Comparisons...

Classical Mechanics

$$\frac{d}{dt}\vec{r}(t) = \frac{\vec{p}(t)}{m} \qquad \frac{d}{dt}\vec{p}(t) = -\nabla V(\vec{r}, t)$$

Quantum Physics

$$\frac{d}{dt} < \vec{r} >= \frac{< \vec{p} >}{m} \qquad \frac{d}{dt} < \vec{p} >=< -\nabla V >$$

Ehrenfest's equations. They are the equations of motion for the mean values.

Expectation values

What about the energy? It may take different values with different probabilities.

Total energy $H = \frac{p^2}{2m} + V(\vec{r}, t)$

$$\int d\vec{r}\, \Psi^*(\vec{r}, t) \frac{(-i\hbar\nabla)^2}{2m} \Psi(\vec{r}, t) \quad \Rightarrow \quad < K >$$

$$\int d\vec{r}\, \Psi^*(\vec{r}, t)\, V(\vec{r}, t)\, \Psi(\vec{r}, t) \quad \Rightarrow \quad < V >$$

Therefore

$$< H >= \int d\vec{r}\, \Psi^*(\vec{r}, t) \left[\frac{(-i\hbar\nabla)^2}{2m} + V(\vec{r}, t) \right] \Psi(\vec{r}, t)$$

Moreover

$$\frac{d}{dt} < H >= \int d\vec{r}\, \Psi^*(\vec{r}, t) \left(\frac{\partial V(\vec{r}, t)}{\partial t} \right) \Psi(\vec{r}, t) \equiv < \frac{\partial V(\vec{r}, t)}{\partial t} >$$

Notice that when dealing with a time-independent potential, i.e. $V(\vec{r})$, $\frac{\partial V}{\partial t} = 0$ and therefore $\frac{d}{dt} < H >= 0$

Summary. 1-D

Classical Physics	Quantum Physics
$\frac{dx(t)}{dt} = \frac{p_x(t)}{m}$	$\frac{d < x >}{dt} = \frac{< p_x >}{m}$
$\frac{dp_x(t)}{dt} = -\frac{\partial V(x,t)}{\partial x}$	$\frac{d < p_x >}{dt} = -< \frac{\partial V(x,t)}{\partial x} >$
$\frac{d\mathcal{H}(x(t),p_x(t),t)}{dt} = \frac{\partial V(x,t)}{\partial t}$	$\frac{d < \mathcal{H} >}{dt} = < \frac{\partial V}{\partial t} >$

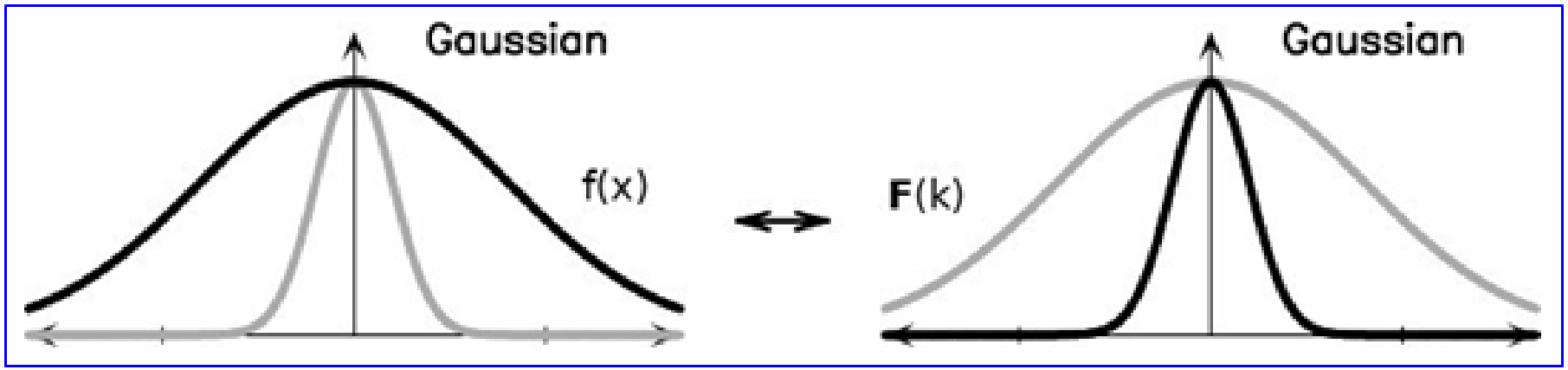
The Fourier transform revisited. A brief reminder

Some properties of the Fourier transform (**in 1-D**):

$$\tilde{f}(k_x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, e^{-ik_x x} f(x)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk_x\, e^{ik_x x} \tilde{f}(k_x)$$

An important property of FT is that if $f(x)$ is a ”narrow” function, then $\tilde{f}(k_x)$ is a ”wide” function, and viceversa.



The Fourier transform revisited. A brief reminder

Some properties of the Fourier transform (in 1-D): Recall that

$$\tilde{\Psi}(k_x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, e^{-ik_x x}\, \Psi(x) \qquad \Psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk_x\, e^{ik_x x}\, \tilde{\Psi}(k_x)$$

In QM, $p_x = \hbar k_x$ and we define the FT:

$$\tilde{\Psi}(p_x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\, e^{-ip_x x/\hbar}\, \Psi(x) \qquad \Psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp_x\, e^{ip_x x/\hbar}\, \tilde{\Psi}(p_x)$$

(therefore $\tilde{\Psi}(k_x) = \hbar^{1/2}\tilde{\Psi}(p_x)$)

Parseval-Plancherel formula:

$$\int_{-\infty}^{\infty} dx\ |\Psi(x)|^2 = \int_{-\infty}^{\infty} dp_x\ \left|\tilde{\Psi}(p_x)\right|^2$$

The Fourier transform revisited. A brief reminder

Fourier transform (3-D):

$$\tilde{\Psi}(k) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int dr\, e^{-ik\cdot r}\, \Psi(r) \qquad \Psi(r) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int dk\, e^{ik\cdot r}\, \tilde{\Psi}(k)$$

$$\left(k \cdot r = k_x x + k_y y + k_z z\right)$$

In QM, $p = \hbar k$ so we will write the FT: (notice $\tilde{\Psi}(k) = \hbar^{3/2}\tilde{\Psi}(p)$)

$$\tilde{\Psi}(p) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} \int dr\, e^{-ip\cdot r/\hbar}\, \Psi(r) \qquad \Psi(r) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} \int dp\, e^{ip\cdot r/\hbar}\, \tilde{\Psi}(p)$$

$$\left(p \cdot r = p_x x + p_y y + p_z z\right)$$

Parseval-Plancherel formula:

$$\int dr\, |\Psi(r)|^2 = \int dp\, \left|\tilde{\Psi}(p)\right|^2$$

The momentum space

The previous expressions for $< r >= (< x >, < y >, < z >)$ look rather intuitive. However, the expression for $< p >= (< p_x >, < p_y >, < p_z >)$ is very different. Is it possible to find another, more intuitive, way of writing it ?

Consider the **momentum space**, which is obtained by means of the Fourier transformation.

Assume that $\Psi(r,t)$ is normalized and evaluate its Fourier transform $\tilde{\Psi}(p,t)$

$$\tilde{\Psi}(p,t) = \frac{1}{(2\pi\hbar)^{3/2}} \int dr\, \Psi(r,t) e^{-ipr/\hbar}$$

$$\Psi(r,t) = \frac{1}{(2\pi\hbar)^{3/2}} \int dp\, \tilde{\Psi}(p,t) e^{ipr/\hbar}$$

$\left|\tilde{\Psi}(p,t)\right|^2 dp$ $\longrightarrow$ Probability that at time *t* the particle has a linear momentum within the volume element *d*p centered around p.

The momentum space

In 1-D. Remember that:

$$\tilde{\Psi}(p_x,t)=\frac{1}{\sqrt{2\pi\hbar}}\int dx\,\Psi(x,t)e^{-ip_x x/\hbar} \qquad \Psi(x,t)=\frac{1}{\sqrt{2\pi\hbar}}\int dp_x\,\tilde{\Psi}(p_x,t)e^{ip_x x/\hbar}$$

If $\Psi(x,t)$ was normalized, then

$$\int dp_x\tilde{\Psi}^*(p_x,t)\tilde{\Psi}(p_x,t)=\int dp_x\left|\tilde{\Psi}(p_x,t)\right|^2=1$$

and now: (notice that $\left|\tilde{\Psi}(p_x,t)\right|^2$ can be considered as a probability density function for a continuous random variable p_x)

$\left|\tilde{\Psi}(p_x,t)\right|^2 dp_x$ ⟶ Probability that at time *t* the particle has a value of the linear momentum within the range (p_x, $p_x + dp_x$).

Moreover

$$< p_x >=\int dp_x\tilde{\Psi}^*(p_x,t)p_x\tilde{\Psi}(p_x,t) \qquad < x >=\int dp_x\tilde{\Psi}^*(p_x,t)\left(i\hbar\frac{\partial}{\partial p_x}\right)\tilde{\Psi}(p_x,t)$$

The momentum space

In 3-D

If $\Psi(r,t)$ was normalized, then

$$\int dp\ \tilde{\Psi}^*(p,t)\tilde{\Psi}(p,t) = \int dp\ \left|\tilde{\Psi}(p,t)\right|^2$$

and now

$$< p_x >= \int dp\ \tilde{\Psi}^*(p,t)p_x\tilde{\Psi}(p,t)$$

$$< p_y >= \int dp\ \tilde{\Psi}^*(p,t)p_y\tilde{\Psi}(p,t)$$

$$< p_z >= \int dp\ \tilde{\Psi}^*(p,t)p_z\tilde{\Psi}(p,t)$$

The momentum space

As for the calculation of $< x >$, $< y >$, $< z >$, we proceed as follows.

Assume $\Psi(r,t)$ is normalized and we obtain its FT, $\tilde{\Psi}(p,t)$. Then

$$< x >= \int dp \tilde{\Psi}^*(p,t) \left(i\hbar \frac{\partial}{\partial p_x} \right) \tilde{\Psi}(p,t)$$

and similar expressions for $< y >$, $< z >$

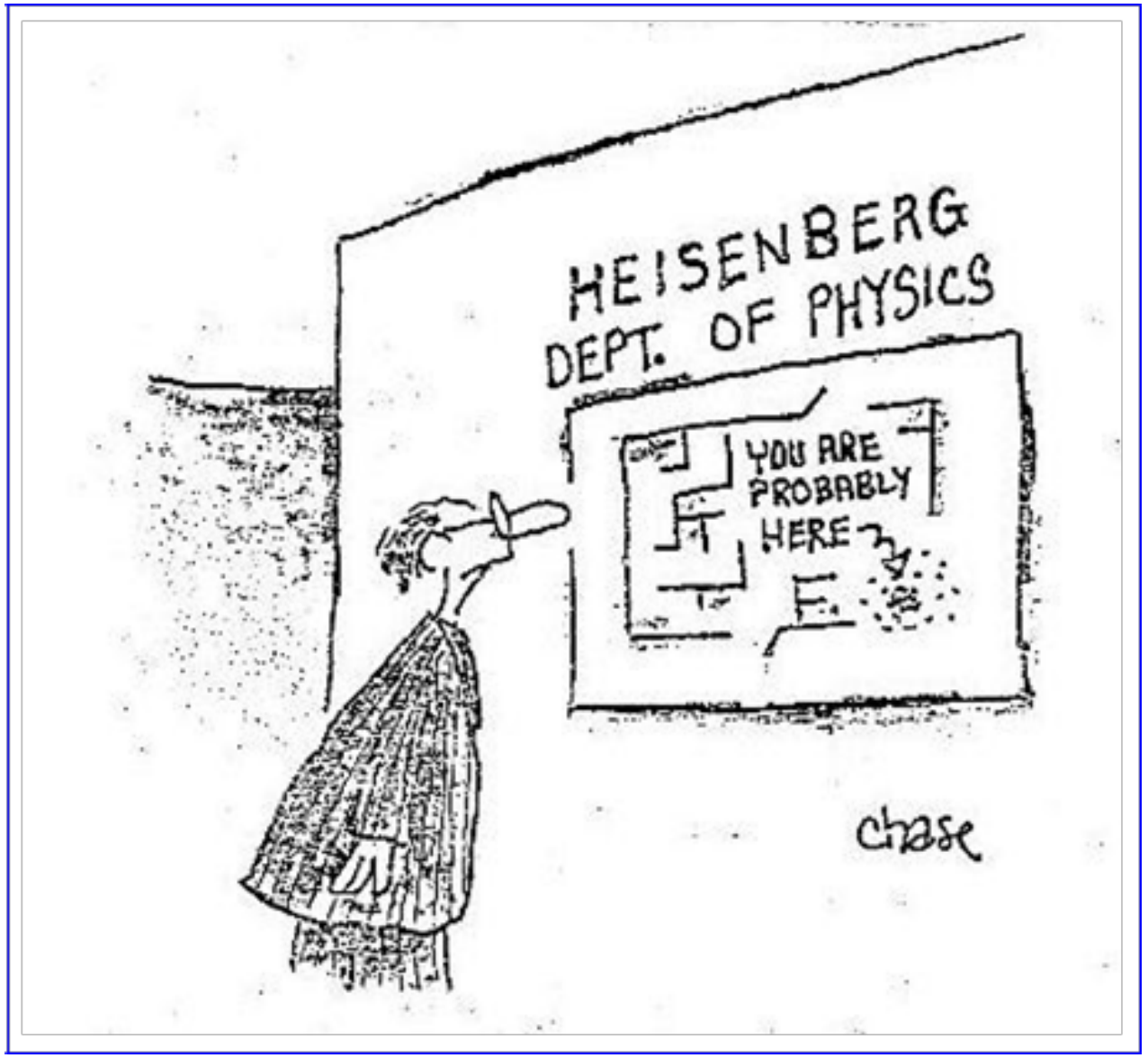
Now, **Schröedinger's equation becomes (3-D)**

$$i\hbar \frac{\partial}{\partial t} \tilde{\Psi}(p,t) = \frac{p^2}{2m} \tilde{\Psi}(p,t) + \frac{1}{(2\pi\hbar)^{3/2}} \int dp' \tilde{V}(p-p') \tilde{\Psi}(p',t)$$

where

$$\tilde{V}(p) = \frac{1}{(2\pi\hbar)^{3/2}} \int dr\, V(r) e^{-ipr/\hbar}$$

Uncertainties



Uncertainties

1-D. Particle characterized by a **normalized** $\Psi(x,t)$. A magnitude that measures the scatter around the mean value is the variance

$$Var(x) = E(x - E(x))^2 =< (x -< x >)^2 >=< x^2 > -< x >^2$$

Standard deviation $\Delta x = (< x^2 > -< x >^2)^{1/2}$

$$\Delta x = \left[\int dx\, \Psi^*(x,t)(x^2 -< x >^2)\Psi(x,t)\right]^{1/2}$$

$$\Delta p_x = (< p_x^2 > -< p_x >^2)^{1/2}$$

$$\Delta p_x = \left[\int dx\, \Psi^*(x,t)\left(\left(-i\hbar\frac{\partial}{\partial x}\right)^2 -< p_x >^2\right)\Psi(x,t)\right]^{1/2}$$

$$\Delta p_x = \left[\int dp_x\, \tilde{\Psi}^*(p_x,t)(p_x^2 -< p_x >^2)\tilde{\Psi}(p_x,t)\right]^{1/2}$$

Uncertainties

3-D. Particle characterized by a normalized $\Psi(r,t)$.

$$Var(x) = E(x - E(x))^2 = < (x - < x >)^2 > = < x^2 > - < x >^2$$

Standard deviation

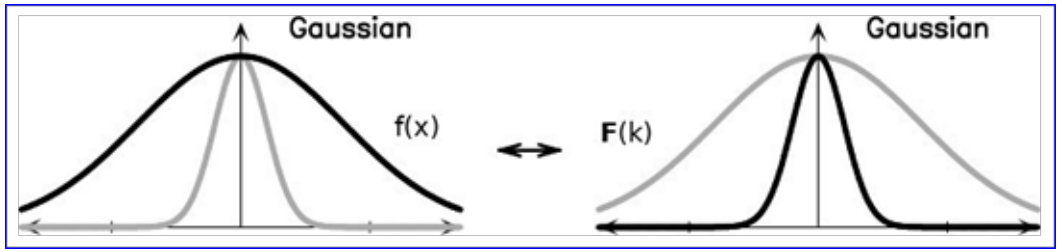
$$\begin{cases} \Delta x = (< x^2 > - < x >^2)^{1/2} \\ \Delta y = (< y^2 > - < y >^2)^{1/2} \\ \Delta z = (< z^2 > - < z >^2)^{1/2} \end{cases}$$

$$\Delta x = \left[\int dr\, \Psi^*(r,t)(x^2 - < x >^2)\Psi(r,t) \right]^{1/2}$$

$$\Delta p_x = \left[\int dr\, \Psi^*(r,t) \left(\left(-i\hbar \frac{\partial}{\partial x} \right)^2 - < p_x >^2 \right) \Psi(r,t) \right]^{1/2}$$

$$\Delta p_x = \left[\int dp\, \tilde{\Psi}^*(p,t)(p_x^2 - < p_x >^2)\tilde{\Psi}(p,t) \right]^{1/2}$$

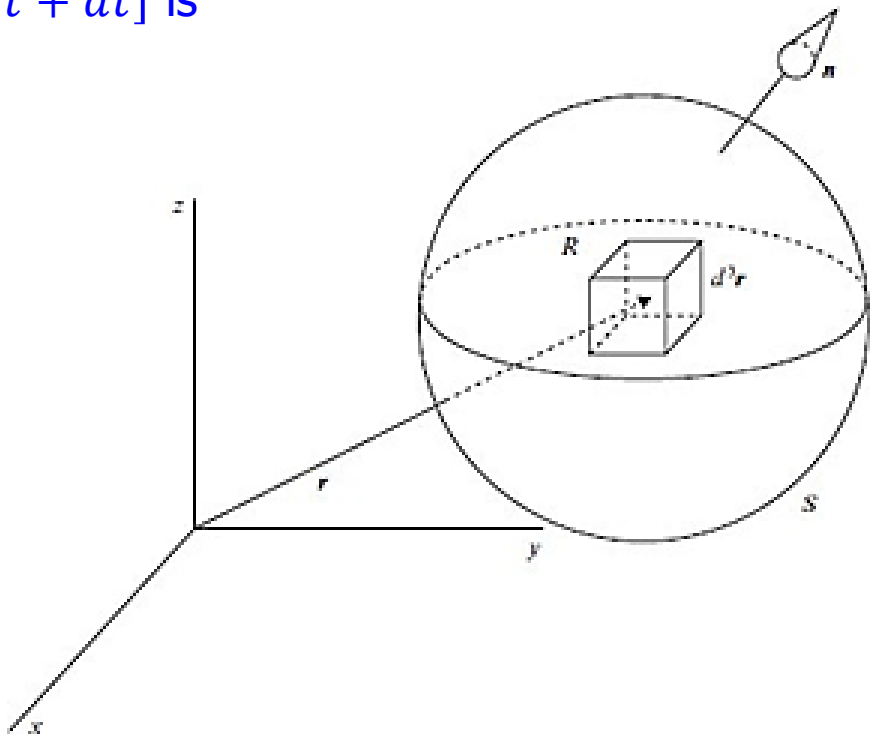
An important property of FT is that if $f(x)$ is a "narrow" function, then $\tilde{f}(k_x)$ is a "wide" function, and viceversa.



Hydrodynamic Image and Probability Current

When the particle moves, the same happens with the associated probability. How does the probability propagate?

Ideas from classical hydrodynamics. Consider a fluid moving in a region: $\rho\,(\,\vec{r}\,,\,t\,)$ the fluid's mass density at point $\vec{r}$ and time t. Let $\mathcal{R}$ represent a volume which is bounded by a surface $\mathcal{S}$. Then, the mass variation inside $\mathcal{R}$ during the time interval $[t, t+dt]$ is



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$$\int_{\mathcal{R}} d\vec{r}\,[\rho(\vec{r},+dt) - \rho(\vec{r},t)] \approx \left[\int_{\mathcal{R}} d\vec{r}\,\frac{\partial\rho(\vec{r},t)}{\partial t}\right] dt$$

This variation is due to the fact that the fluid comes in/comes out through the surface $\mathcal{S}$.

Notice that the integral is > 0 if the amount fluid that comes in > comes out, and viceversa.

Hydrodynamic Image and Probability Current

Introduce $\vec{j}(\vec{r},t) \Rightarrow$ (mass) current density of the fluid:

It's defined such that its flux through $\mathcal{S}$ equals the net amount of fluid (comes out - comes in) that crosses $\mathcal{S}$ per unit time. ($\vec{n}$ is an external unit vector on $\mathcal{S}$)

$$\int_{\mathcal{S}} \vec{n}\,\vec{j}(\vec{r},t)\,dS = -\int_{\mathcal{R}} d\vec{r}\,\frac{\partial\rho(\vec{r},t)}{\partial t} = \int_{\mathcal{R}} d\vec{r}\,\nabla\,\vec{j}(\vec{r},t)$$

$$\int_{\mathcal{R}} d\vec{r}\left[\frac{\partial\rho(\vec{r},t)}{\partial t} + \nabla\,\vec{j}(\vec{r},t)\right] = 0$$

and by taking $\mathcal{R} \equiv d\vec{r}$, we get ....

Continuity equation for a fluid (conservation of the total amount of mass)

$$\boxed{\frac{\partial\rho(\vec{r},t)}{\partial t} = -\nabla\,\vec{j}(\vec{r},t)}$$

Hydrodynamic Image and Probability Current

We return to our microscopic particle. How does its probability change in space and time? Apply the previous ideas to the probability fluid. $\Psi(\vec{r},t)$ is the wave function of a particle moving in a real potential $V(\vec{r},t) = V^*(\vec{r},t)$

$$-\frac{\hbar^2}{2m}\Delta\Psi(\vec{r},t) + V(\vec{r},t)\Psi(\vec{r},t) = i\hbar\frac{\partial\Psi(\vec{r},t)}{\partial t}$$

$$-\frac{\hbar^2}{2m}\Delta\Psi^*(\vec{r},t) + V^*(\vec{r},t)\Psi^*(\vec{r},t) = -i\hbar\frac{\partial\Psi^*(\vec{r},t)}{\partial t}$$

Analyze the time evolution of $\Psi(\vec{r},t)\Psi^*(\vec{r},t) = |\Psi(\vec{r},t)|^2$

$$\frac{\partial}{\partial t}\Psi(\vec{r},t)\Psi^*(r,t) = \frac{\partial\Psi^*}{\partial t}\Psi + \Psi^*\frac{\partial\Psi}{\partial t} = \frac{i\hbar}{2m}[\Psi^*\Delta\Psi - \Psi\Delta\Psi^*]$$

Introduce a **probability current density**

$$\vec{J}(\vec{r},t) = \frac{i\hbar}{2m}[\Psi\nabla\Psi^* - \Psi^*\nabla\Psi]$$

$$\nabla\vec{J} = \frac{i\hbar}{2m}[\Psi\Delta\Psi^* - \Psi^*\Delta\Psi] = -\frac{\partial}{\partial t}|\Psi|^2$$

which looks like a continuity equation, with $|\Psi|^2 \equiv \Psi^*\Psi$ resembling a density.

Hydrodynamic Image and Probability Current

Consider a volume $\mathcal{R}$ bounded by a surface $\mathcal{S}$

$$\frac{\partial}{\partial t}\int_{\mathcal{R}} d\vec{r}|\Psi|^2 = -\int_{\mathcal{R}} d\vec{r}\nabla\vec{j}(\vec{r},t) = -\oint_{\mathcal{S}} dS\,\vec{n}\cdot\vec{J}$$

The probability moves and propagates in space and time in the same way as a fluid.

$\oint_{\mathcal{S}} dS\,\vec{n}\cdot\vec{J}$ $\Rightarrow$ is the net probability flux (comes out - comes in) through the surface $\mathcal{S}$, per unit time.

Suppose $\mathcal{R}$ represents **all space**. By the normalization condition, we know that $\Psi(\vec{r},t)\to 0$ when $r\to\infty$, and therefore $\vec{J}(\vec{r},t)\to 0$. Consequently

$$\oint_{\mathcal{S}_\infty} dS\,\vec{n}\cdot\vec{J}\to 0 \qquad\qquad \frac{\partial}{\partial t}\int_{\mathcal{R}} d\vec{r}|\Psi|^2 = 0$$

The integral is time-independent. Therefore, if it is normalized for a given $t=t_0$ then it will remain normalized for latter times. **(VERY IMPORTANT RESULT !!!!!)**

The probability propagates in space and time in a similar way as a fluid

The time-independent Schrödinger's equation

We have the partial differential equation (1-D):

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2}+V(x,t)\Psi(x,t)=i\hbar\frac{\partial\Psi}{\partial t}$$

A standard technique is searching for solutions in the form of products of functions, i.e. $\Psi(x,t)=\psi(x)\phi(t)$.
Moreover, we assume that the potential energy does not explicitly depend on time t, so $V(x,t)\equiv V(x)$

Substituting into Schrödinger's equation:

$$-\frac{\hbar^2}{2m}\,\phi(t)\frac{d^2\psi(x)}{dx^2}+V(x)\,\psi(x)\,\phi(t)=i\hbar\,\psi(x)\frac{d\phi(t)}{dt}$$

Dividing by $\Psi(x,t)=\psi(x)\phi(t)$

$$\underbrace{-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2}+V(x)}_{\text{It only depends on x}}=\underbrace{i\hbar\,\frac{1}{\phi(t)}\frac{d\phi(t)}{dt}}_{\text{It only depends on t}}$$

The time-independent Schröedinger's equation

Both terms must have a common **constant** value, *G*

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\,\psi(x) = G\,\psi(x)$$

$$\frac{d\phi(t)}{dt} = -i\frac{G}{\hbar}\phi(t)$$

The **first** equation gives solutions which depend on the specific $V(x)$, i.e. on the particular system under study.

The **second** equation has solutions which are independent of the system (remember that the potential is time-independent). In fact

$$\phi(t) = Ae^{-iGt/\hbar}$$

What is the physical meaning of (separation constant) G?.

The time-independent Schrödinger's equation

$$\phi(t) = \cos(Gt/\hbar) - i\sin(Gt/\hbar) = \cos 2\pi(Gt/h) - i\sin 2\pi(Gt/h)$$

Oscillatory function with frequency $\nu = G/h$

De Broglie-Einstein $\nu = E/h$ Therefore $G \equiv E$

Finally:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\,\psi(x) = E\,\psi(x)$$

is the (1-D) **time-independent Schrödinger's equation**

and then

$$\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$$

The time-independent Schrödinger's equation

The time-independent Schrödinger's equation can be written as:

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)\right]\psi(x)=E\,\psi(x)$$

where

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)=H$$

is an operator defined on a space of functions.

An equation such as

$$H\psi(x)=E\psi(x)$$

is known as an eigenvalue equation for *H*, the functions $\psi(x)$ which are solutions of the equation are called the **eigenfunctions** of *H* and the numbers, *E*, are the **eigenvalues** of *H*.

The operator *H*, is called the **hamiltonian** of the system.

Stationary states

In general (3-D):

When the potential is time-independent, i.e. $V(\mathrm{r}, t) = V(\mathrm{r})$ then

$$\Psi(r,t) = \psi(r)\phi(t)$$

$$-\frac{\hbar^2}{2m}\Delta\psi(r) + V(r)\psi(r) = E\psi(r) \qquad\qquad i\hbar\frac{d\phi(t)}{dt} = \mathrm{E}\phi(t)$$

$$\Psi(r,t) = \psi(r)e^{-iEt/\hbar} \longrightarrow \text{is the wave function}$$

When the system is in a state characterized by the previous wave function, $\Psi(r,t)$, then we say that the system is in a **stationary state** with energy E

The time-independent Schrödinger equation

$$H\psi(x) = E\psi(x)$$

is an eigenvalue equation

Now, $\psi(r)$ is an **eigenfunction** and *E* is an **eigenvalue**.

Stationary states

Properties

- All eigenvalues are real numbers
- If $\psi_1(r)$ and $\psi_2(r)$ are eigenfunctions with respective eigenvalues E_1 and E_2 $(E_1 \neq E_2)$ then $\int dr\, \psi_1^*(r)\, \psi_2(r) = 0$ (it is said that $\psi_1(r)$ and $\psi_2(r)$ are orthogonal)
- *E* is a degenerate eigenvalue when it has several different eigenfunctions

Given a *V*(r), the set of all the eigenvalues of *H* is called its spectrum. In general, it consists of:

(i) A discrete set of real values $(E_1, E_2, E_3, \ldots)$. Discrete spectrum

(ii) A continuous set. Continuous spectrum.

Stationary states: Properties

If $\Psi(r,t)$ is a **stationary state**, then:

- $|\Psi(r,t)|^2$ is time-independent
- The associated probability current density is time-independent $\vec{J}(r,t) = \vec{J}(r)$ and $\nabla\vec{J}(r) = 0$

In 1-D we have $\frac{dJ(x)}{dx} = 0 \quad \Longrightarrow \quad J(x) = constant$

- If *E* is the eigenvalue associated to $\Psi(r,t)$, then $< H >= E$

$$< H >= \int dr\, \Psi^*(r,t)\, H\, \Psi(r,t) = E \int dr\, \psi^*(r)\psi(r) = E$$

- $\Delta E = 0$. (The uncertainty is zero)

$$(\Delta E)^2 = \int dr\, \Psi^*(r,t)(H^2 - < H >^2)\Psi(r,t) = 0$$

Taking into account that $\Delta E \Delta t \geq \hbar/2 \quad \Longrightarrow \quad \Delta t \longrightarrow \infty$

- Consider all stationary states: $\psi_{\alpha,\beta}(r)e^{-iE_\alpha t/\hbar}$ (α accounts for all the eigenvalues and β accounts for the degeneracy (for a given α)). Then

$$\Psi(r,t) = \sum_\alpha \sum_\beta c_{\alpha,\beta}\, \psi_{\alpha,\beta}(r) e^{-iE_\alpha t/\hbar}$$

(*Most general solution for a purely discrete spectrum. It is* NOT *a stationary state* !!!)

Required properties of eigenfunctions

We are going to solve the time-independent Schrödinger equation. Some conditions on the solutions:

The eigenfunction $\psi(x)$ and its derivative $d\psi(x)/dx$ must be (for finite, non-singular potentials):

- FINITE. If $\psi(x)$ or $d\psi(x)/dx$ were not finite, the calculation of mean values might lead to non finite results.
- SINGLE VALUED. If $\psi(x)$ or $d\psi(x)/dx$ were not single valued, the calculation of mean values might give different results.
- CONTINUOUS. If $\psi(x)$ were not continuous at a point, then $d\psi(x)/dx$ would not be finite at that point.
- CONTINUOUS. $d\psi(x)/dx$ must be continuous. According to

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\,\psi(x)$$

for finite $V(x)$, E and $\psi(x)$, the $\frac{d^2\psi(x)}{dx^2}$ must be finite $\Rightarrow$ $d\psi(x)/dx$ must be continuous.

Energy quantization in the Schrödinger theory

Acceptable solutions of SE by using qualitative arguments concerning curvatures and slopes of the possible solutions.

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\,\psi(x)$$

The properties of the SE depend on the form of the potential $V(x)$. Let us choose a form for $V(x)$

This specific $V(x)$ may represent the potential energy of an atom that is bound to a similar atom and form a diatomic molecule.

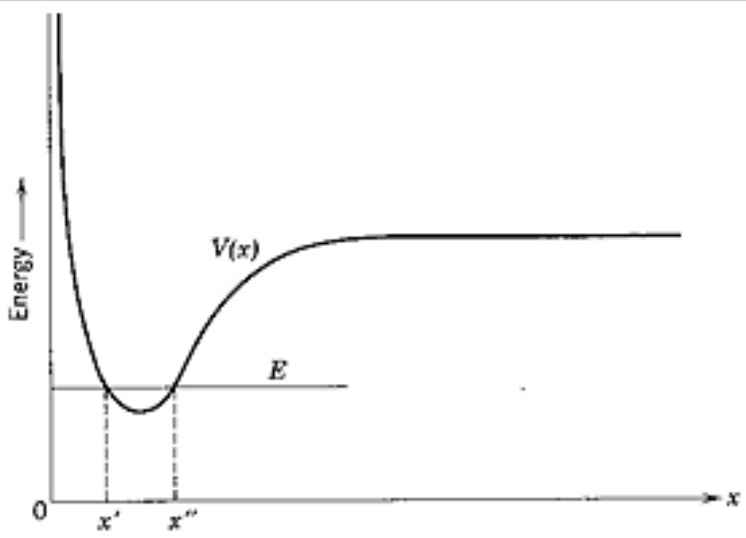


Figure 5-10 The potential energy $V(x)$ used in qualitative arguments concerning the solutions to the time-independent Schroedinger equation, and the total energy E chosen for these arguments.

Energy quantization in the Schrödinger theory

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\,\psi(x)$$

Study the value of $\frac{d^2\psi(x)}{dx^2}$ in terms of $\frac{2m}{\hbar^2}[V(x) - E]$ and $\psi(x)$

Choose an energy E. There are three regions

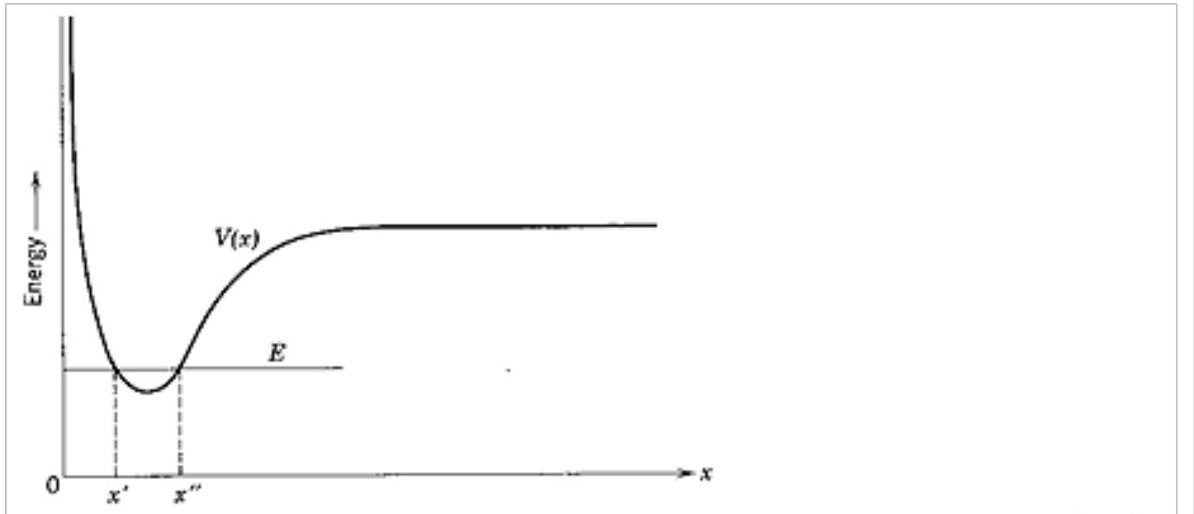


Figure 5-10 The potential energy $V(x)$ used in qualitative arguments concerning the solutions to the time-independent Schroedinger equation, and the total energy E chosen for these arguments.

(i) $x < x' \implies [V(x) - E] > 0 \implies \frac{d^2\psi(x)}{dx^2}$ and $\psi(x)$ have the same sign

(ii) $x' < x < x'' \implies [V(x) - E] < 0 \implies \frac{d^2\psi(x)}{dx^2}$ and $\psi(x)$ have opposite sign

(iii) $x > x'' \implies [V(x) - E] > 0 \implies \frac{d^2\psi(x)}{dx^2}$ and $\psi(x)$ have the same sign

Energy quantization in the Schrödinger theory

Therefore, we have the following relation between the sign of $\psi(x)$ and the sign of $\frac{d^2\psi(x)}{dx^2}$ in the regions defined by the sign of $[V(x) - E]$.

For a given $V(x)$, the time-independent SE establishes a relation between $\psi(x)$ and $\frac{d^2\psi(x)}{dx^2}$ that determines the general behavior of $\psi(x)$.

If we now specify the values at a given x_0 , e.g. $\psi(x_0)$ and $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$ then a **particular** $\psi(x)$ is obtained $\forall x$.

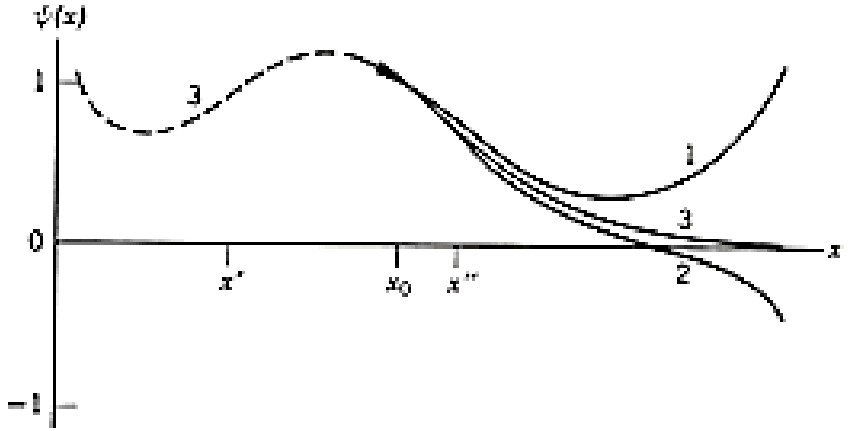
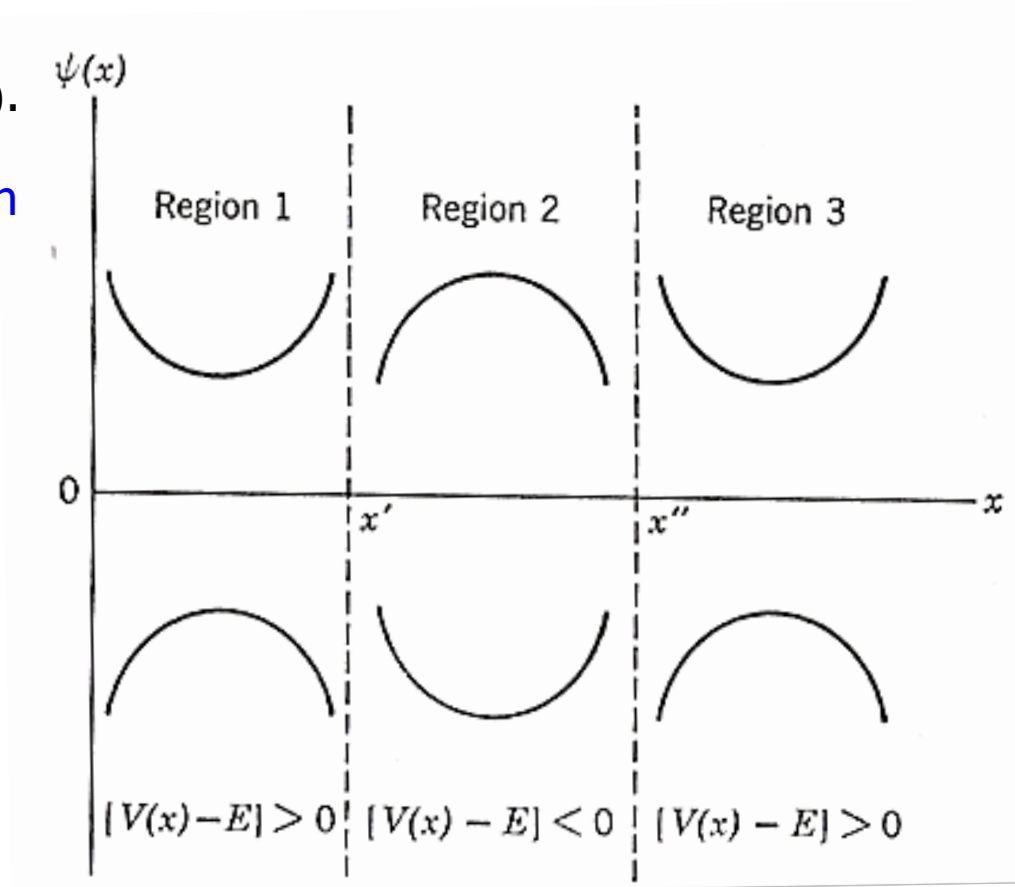


Figure 5-14 Three attempts at finding an acceptable solution to a time-independent Schroedinger equation for an assumed value of the total energy E. The first two (1, 2) failed because the solution became infinite at large x. The third (3) gave the solution with acceptable behavior at large x, but failed because the solution became infinite at small x (dashed curve).



Energy quantization in the Schrödinger theory

CURVE 1. Start integration at x_0 with $\psi(x_0) > 0$ and $\left.\frac{d\psi(x)}{dx}\right|_{x_0} < 0$ and continue the integration towards $x > x_0$.

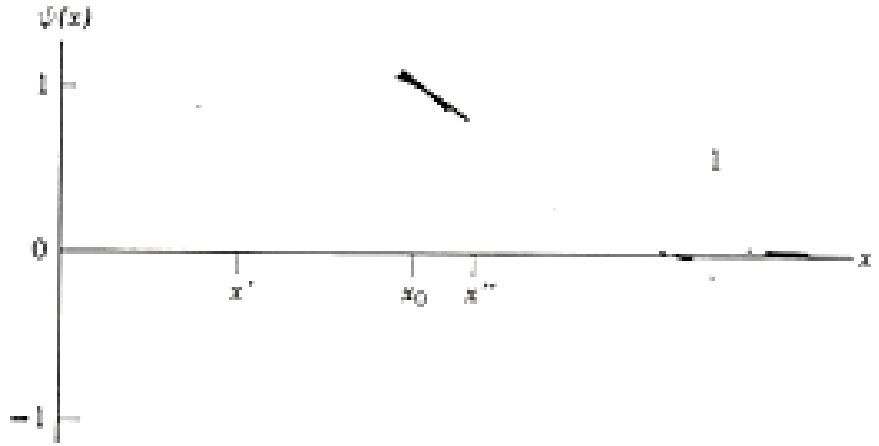


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In Region II, the curve downwards $\psi(x)$ is concave down. and remains so until it enters Region III, $(x > x'')$

Energy quantization in the Schrödinger theory

CURVE 1. Start integration at x_0 with $\psi(x_0) > 0$ and $\left.\frac{d\psi(x)}{dx}\right|_{x_0} < 0$ and continue the integration towards $x > x_0$.

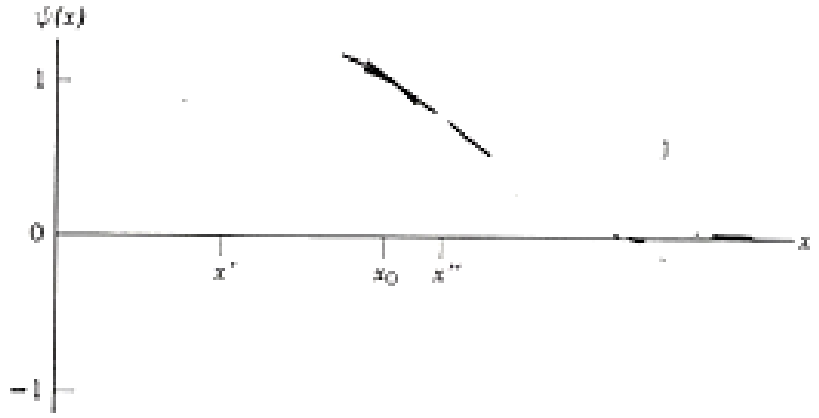


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In Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ was still positive at x'', then $\psi(x)$ will be a …

Energy quantization in the Schrödinger theory

CURVE 1. Start integration at x_0 with $\psi(x_0) > 0$ and $\left.\frac{d\psi(x)}{dx}\right|_{x_0} < 0$ and continue the integration towards $x > x_0$.

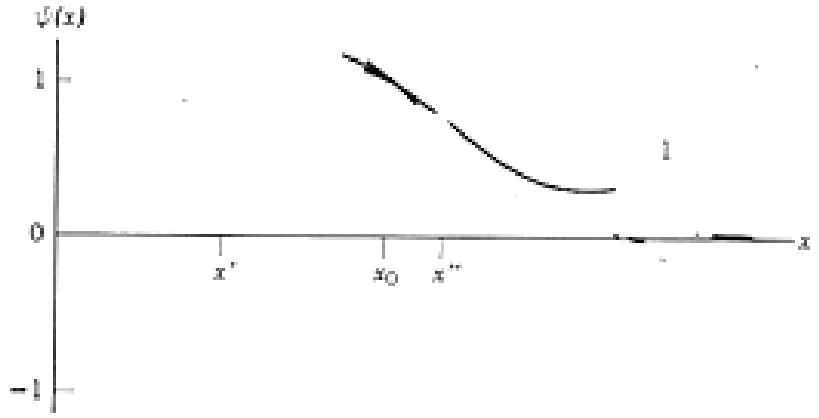


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In Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ was still positive at x'', then $\psi(x)$ will be a concave upwards curve at Region III.

Energy quantization in the Schrödinger theory

CURVE 1. Start integration at x_0 with $\psi(x_0) > 0$ and $\left.\frac{d\psi(x)}{dx}\right|_{x_0} < 0$ and continue the integration towards $x > x_0$.

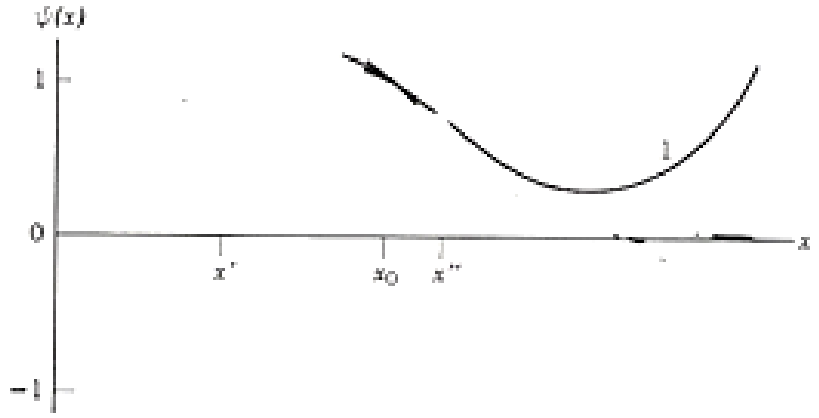


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In Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ was still positive at x'', then $\psi(x)$ will be a concave upwards curve at Region III. Now, it will increase in value and $\rightarrow \infty$

Energy quantization in the Schrödinger theory

CURVE 2. Start integration at x_0 with the same $\psi(x_0)(> 0)$ as before and $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$ more negative than before.

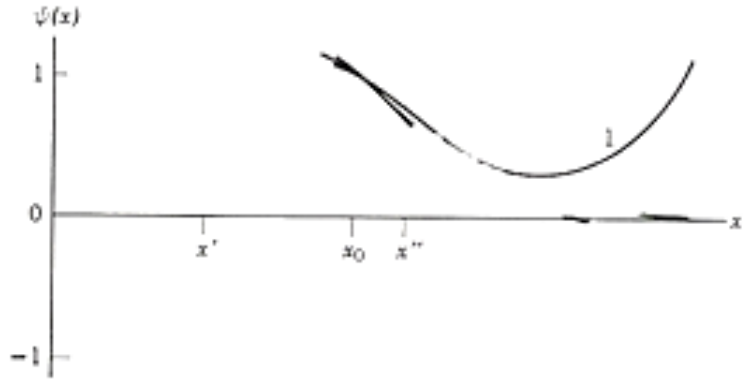


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In Region II, the curve $\psi(x)$ is again concave downwards

Energy quantization in the Schrödinger theory

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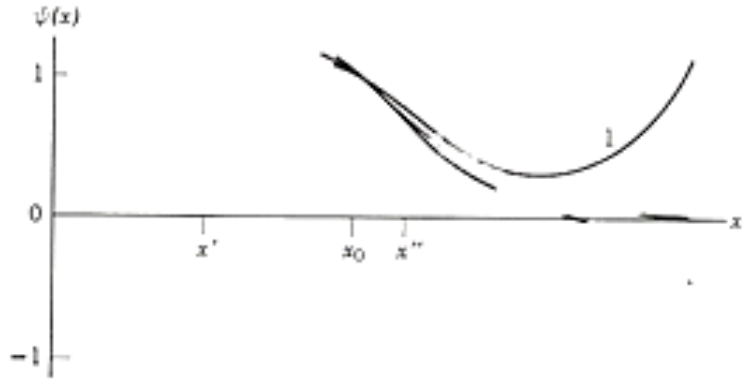


Figure 5-14 Three attempts at finding an acceptable solution to a time-independent Schroedinger equation for an assumed value of the total energy E. The first two (1, 2) failed because the solution became infinite at large x. The third (3) gave the solution with acceptable behavior at large x, but failed because the solution became infinite at small x (dashed curve).

In Region II, the curve $\psi(x)$ is again concave until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

Energy quantization in the Schrödinger theory

CURVE 2. Start integration at x_0 with the same $\psi(x_0)(> 0)$ as before and $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$ more negative than before.

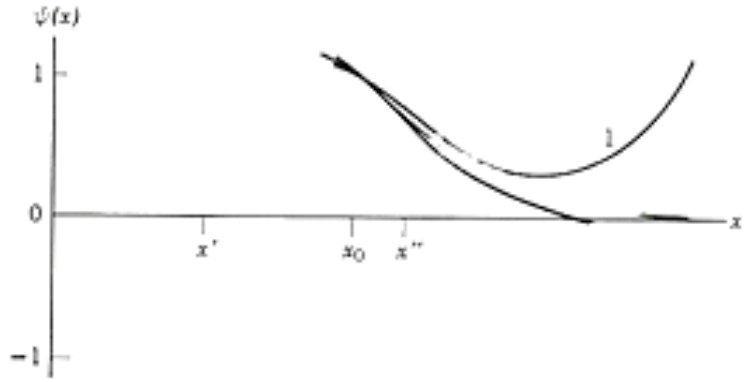


Figure 5-14 Three attempts at finding an acceptable solution to a time-independent Schroedinger equation for an assumed value of the total energy *E*. The first two (1, 2) failed because the solution became infinite at large *x*. The third (3) gave the solution with acceptable behavior at large *x*, but failed because the solution became infinite at small *x* (dashed curve).

In Region II, the curve $\psi(x)$ is again concave until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ is still positive, then $\psi(x)$ will be a concave upwards curve at Region III but it crosses the x-axis and $\psi(x) < 0$.

Energy quantization in the Schrödinger theory

CURVE 2. Start integration at x_0 with the same $\psi(x_0)(> 0)$ as before and $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$ more negative than before.

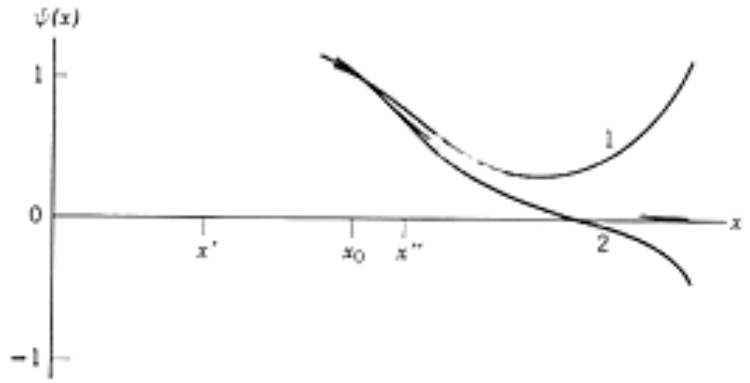


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In Region II, the curve $\psi(x)$ is again concave until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ is still positive, then $\psi(x)$ will be a concave up. curve at Region III but it crosses the *x*-axis and $\psi(x) < 0$.

Then the curve becomes concave downwards and goes $\rightarrow -\infty$

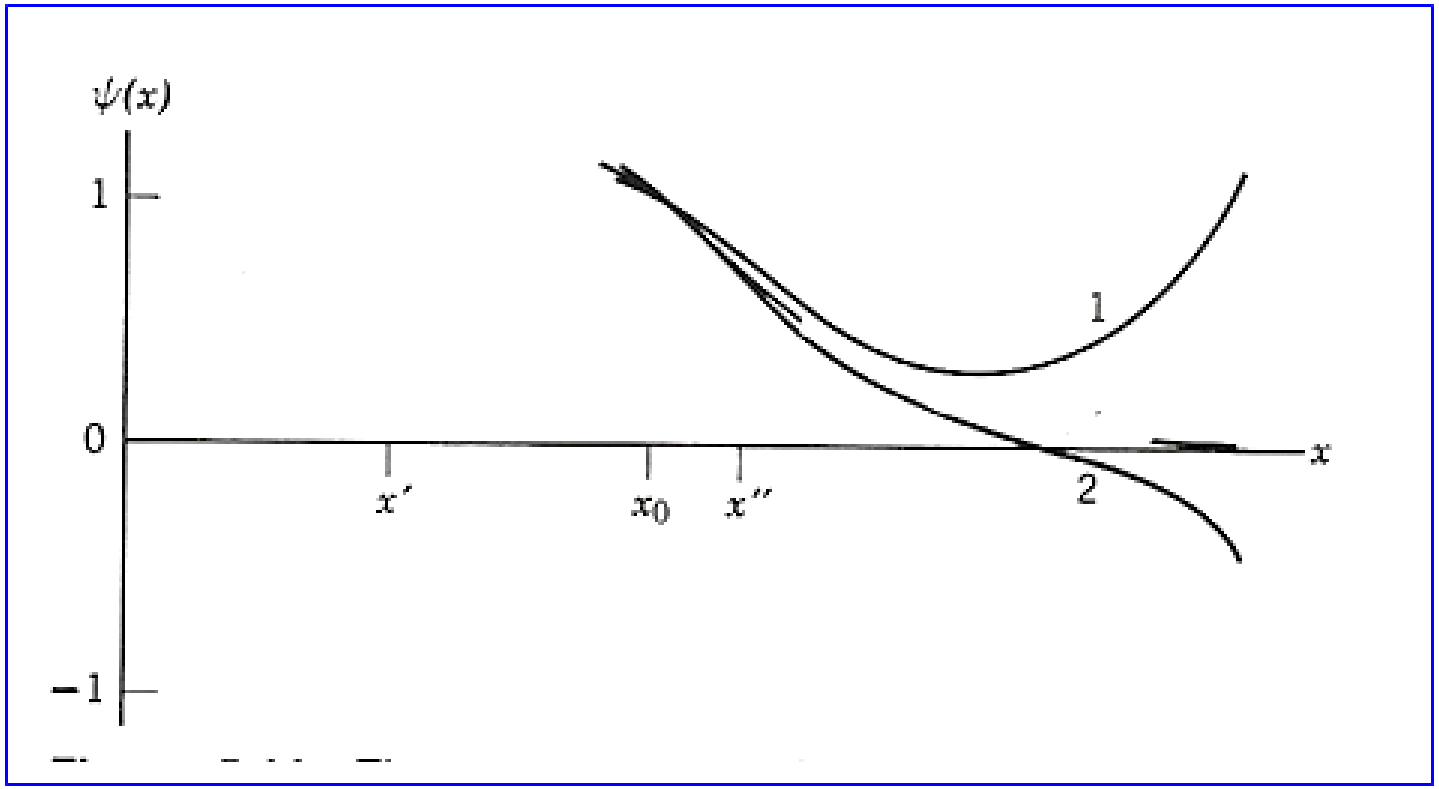
Energy quantization in the Schrödinger theory

THINGS ARE MORE COMPLICATED THAN YOU THOUGHT !!

LET'S HAVE ANOTHER GO !!

Energy quantization in the Schrödinger theory

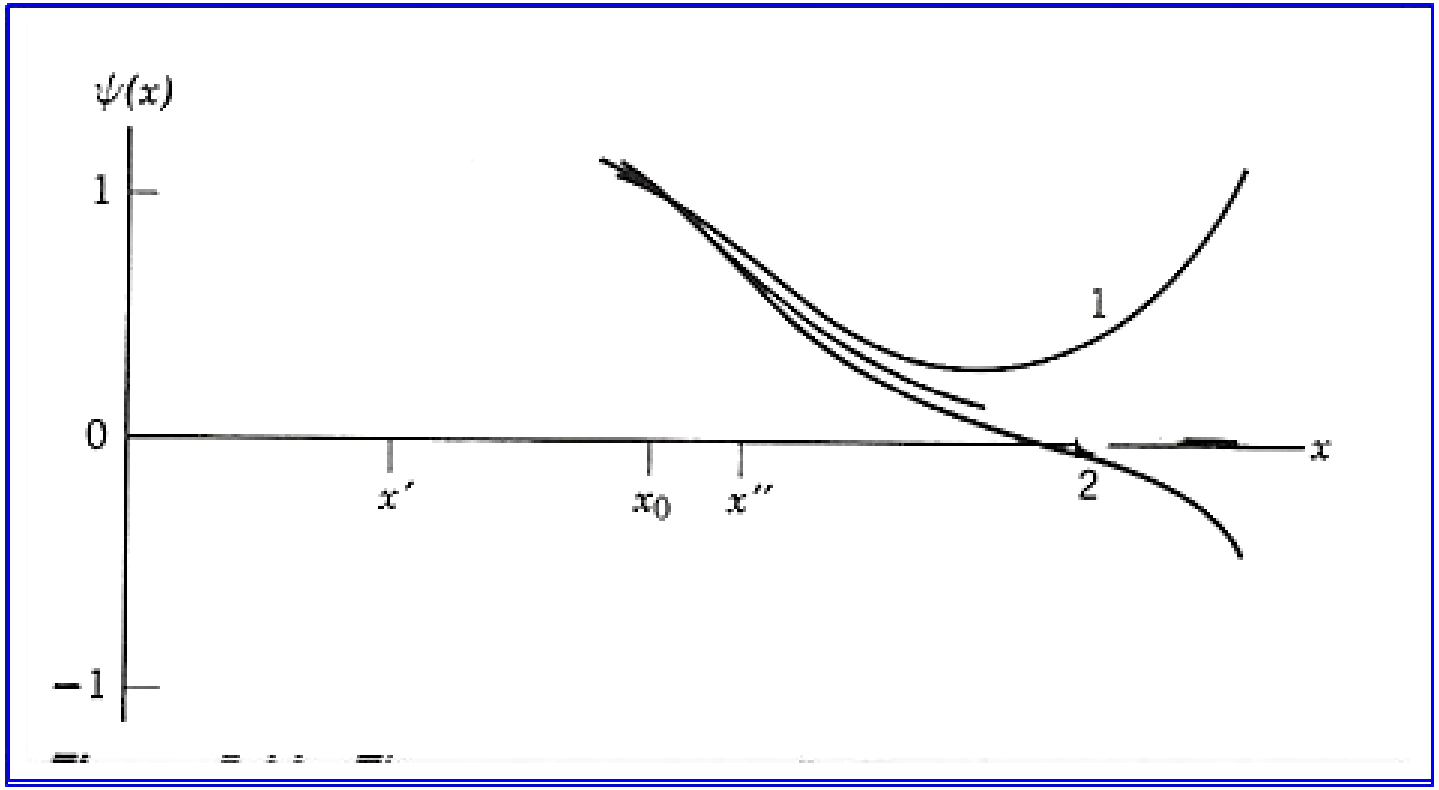
CURVE 3. Start integration at x_0 with the same $\psi(x_0)(> 0)$ as before and an appropriate $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$



In Region II, the curve $\psi(x)$ is again concave down. until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign.

Energy quantization in the Schrödinger theory

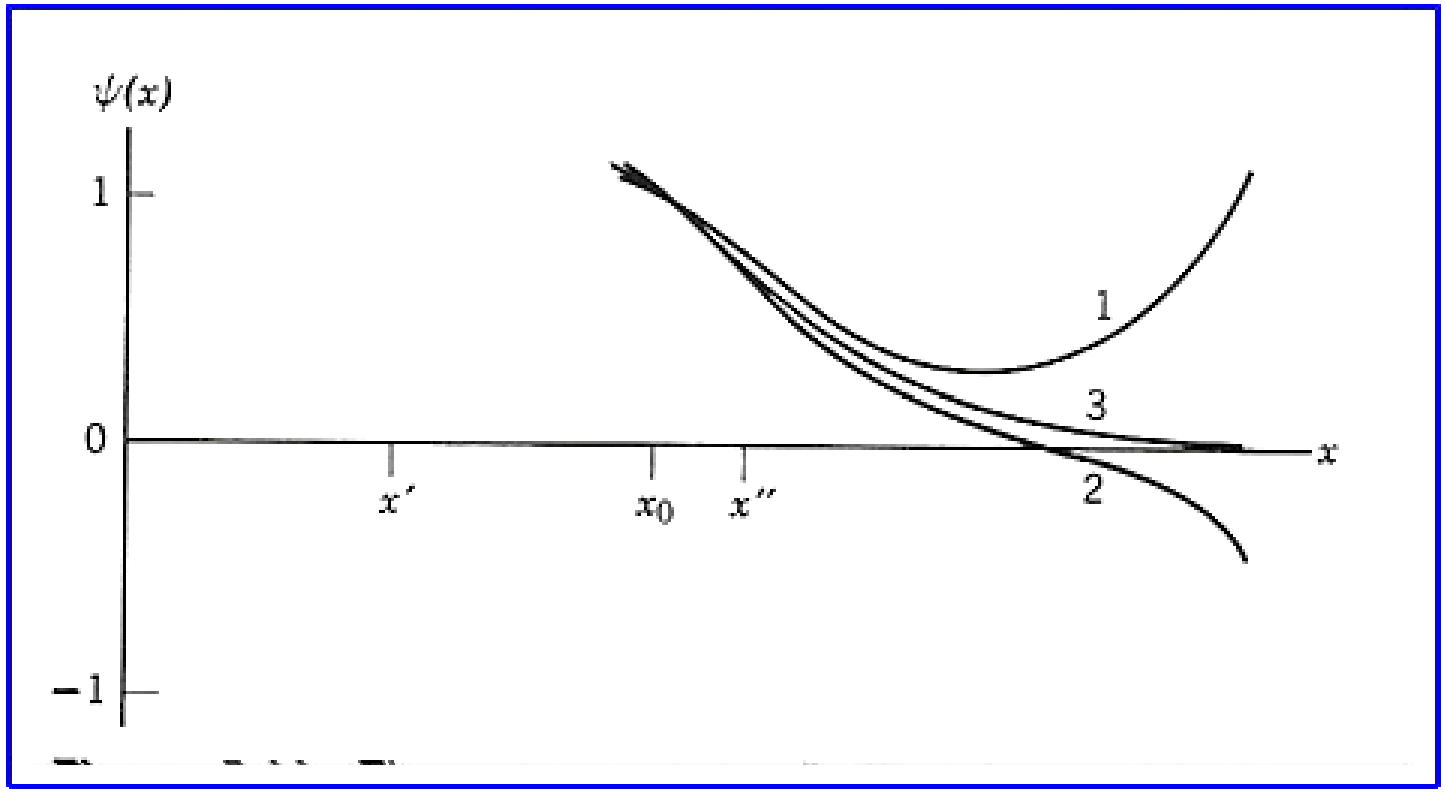
CURVE 3. Start integration at x_0 with the same $\psi(x_0)(> 0)$ as before and an appropriate $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$



In Region II, the curve $\psi(x)$ is again concave d. until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign. If $\psi(x)$ is still positive, then $\psi(x)$ will be a concave up. curve

Energy quantization in the Schrödinger theory

CURVE 3. Start integration at x_0 with the same $\psi(x_0)(>0)$ as before and an appropriate $\left.\frac{d\psi(x)}{dx}\right|_{x_0}$



In Region II, the curve $\psi(x)$ is again concave d. until it enters Region III, $(x > x'')$, where $[V(x) - E]$ changes sign. If $\psi(x)$ is still positive, then $\psi(x)$ will be a concave up. curve at Region III and the $\psi(x) \to 0$ when $x \to \infty$

Energy quantization in the Schrödinger theory

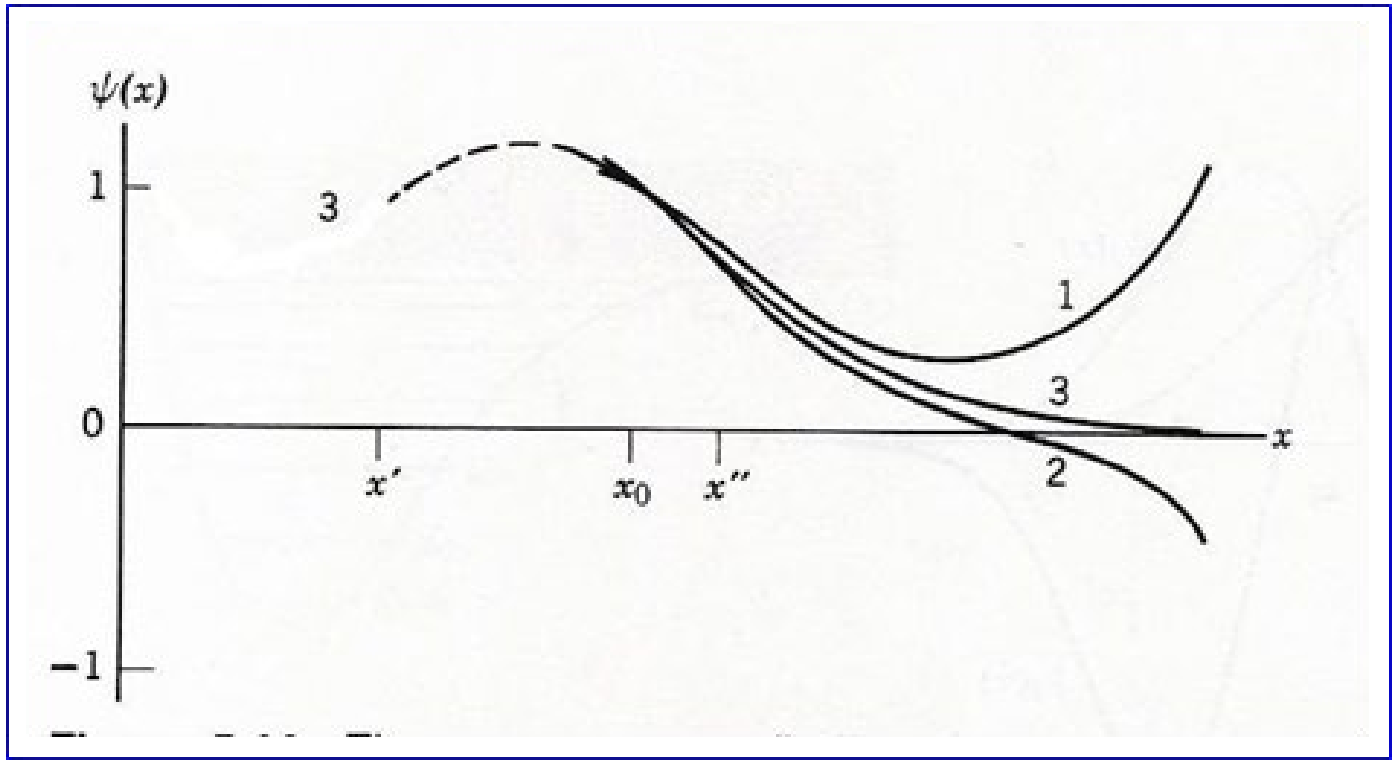
THINGS ARE MORE COMPLICATED THAN YOU THOUGHT !!

WAIT A MINUTE. !!

WE HAVE NOT FINISHED YET !!

Energy quantization in the Schrödinger theory

But now, we have to continue the integration towards $x = 0$.



In Region II, the curve $\psi(x)$ is concave d. until it enters Region I, $(x < x')$, where $[V(x) - E]$ changes sign.

If $\psi(x)$ remains positive, then $\psi(x)$ will be a concave up. curve and likely $\psi(x) \to \infty$ when $x \to 0$.

If $\psi(x)$ becomes negative then $\psi(x)$ will be a concave d. curve and likely $\psi(x) \to -\infty$ when $x \to 0$

Energy quantization in the Schrödinger theory

The conclusion from the previous calculation is that **for that specific *E* value**, there is no physically acceptable solution satisfying the required boundary conditions and normalizability.

By repeating this procedure for many different choices of the energy E, it will, eventually, be possible to find a value E_1 for which the SE has an acceptable solution $\psi_1(x)$.

In general, there will be a number of allowed values of total energy, $E_1, E_2, E_3, \ldots$ for which the SE has acceptable solutions $\psi_1(x), \psi_2(x), \psi_3(x), \ldots$

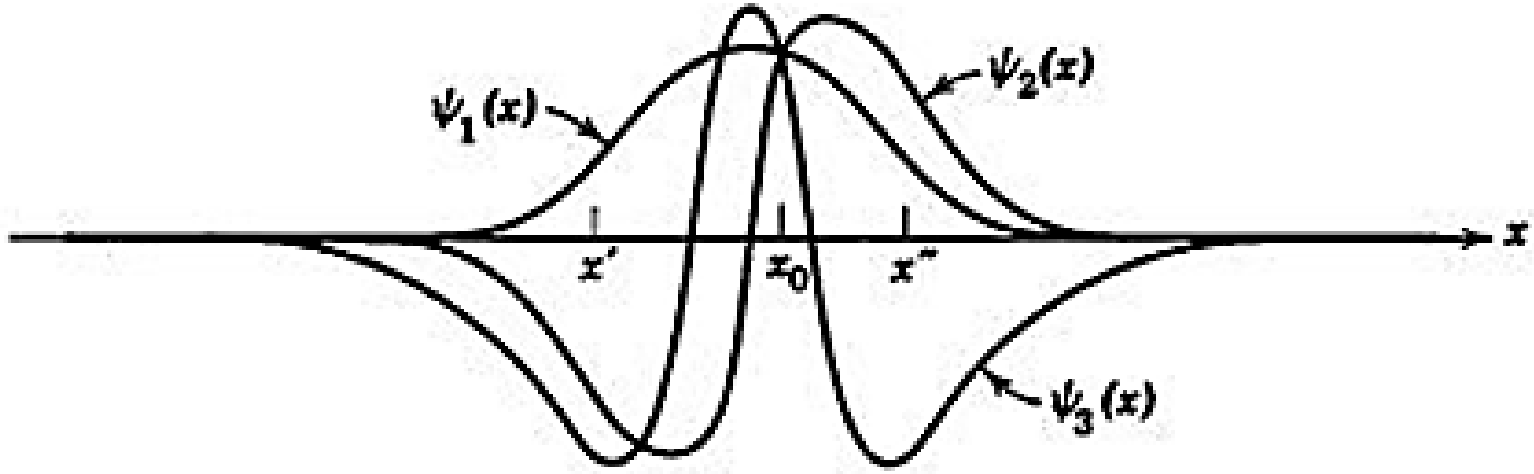


Figure 7–4. The first three acceptable solutions of a typical time independent Schroedinger equation.

Energy quantization in the Schrödinger theory

Notice that the behavior of $\psi_1(x)$ for both small and large x is the same as the behavior of the function shown in curve 3 for large x.

For $x < x_0$, the behavior of $\psi_2(x)$ is at first similar to $\psi_1(x)$, but since its second derivative is larger in magnitude, $\psi_2(x)$ crosses the x-axis at some value $< x_0$ but greater than x'. Now the function becomes concave up. but at x' becomes concave down. and then $\psi_1(x \longrightarrow 0) \longrightarrow 0$.

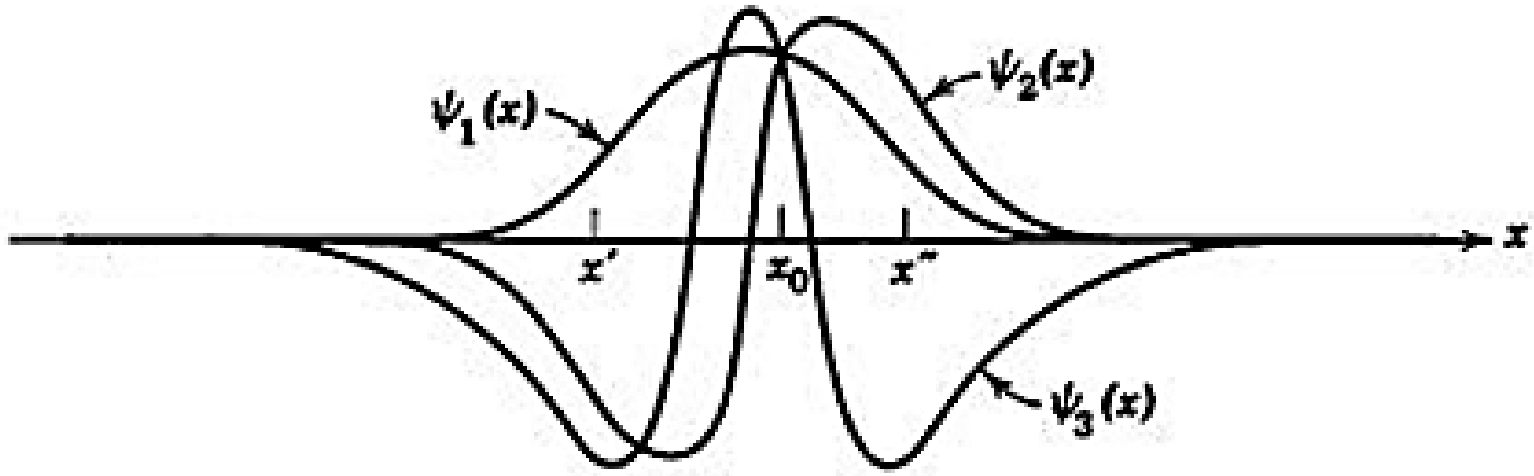


Figure 7–4. The first three acceptable solutions of a typical time independent Schroedinger equation.

Energy quantization in the Schrödinger theory

Moreover, $E_2 > E_1$

Consider x_0, where $\psi_1(x_0) = \psi_2(x_0)$. According to the figure

$$\left|\frac{d^2\psi_2(x)}{dx^2}\right|_{x_0} > \left|\frac{d^2\psi_1(x)}{dx^2}\right|_{x_0}$$

and from the SE we get $|V(x_0) - E_2| > |V(x_0) - E_1|$, so if this holds at x_0 then $E_2 > E_1$ (since $E > V(x)$).

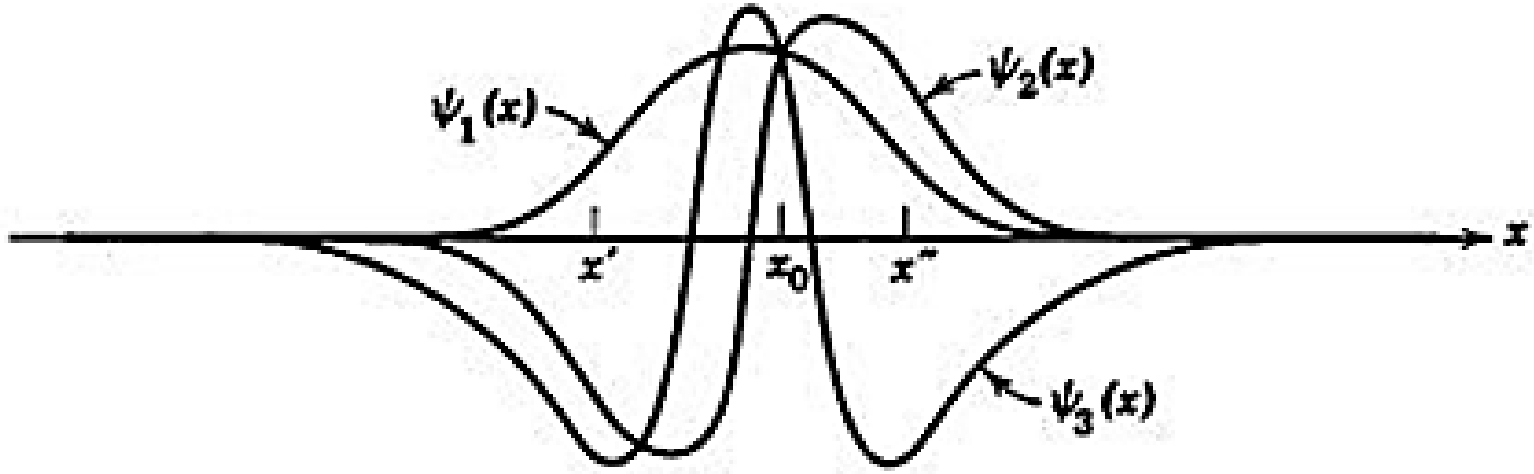


Figure 7–4. The first three acceptable solutions of a typical time independent Schroedinger equation.

Energy quantization in the Schrödinger theory

From similar arguments it can be shown that $E_1 < E_2 < E_3 <....$

When the relation between the total energy of a particle and its potential energy is such that classically the particle would be bound to a limited region of space because the potential energy would exceed the total energy outside the region, the SE predicts that the total energy is quantized.

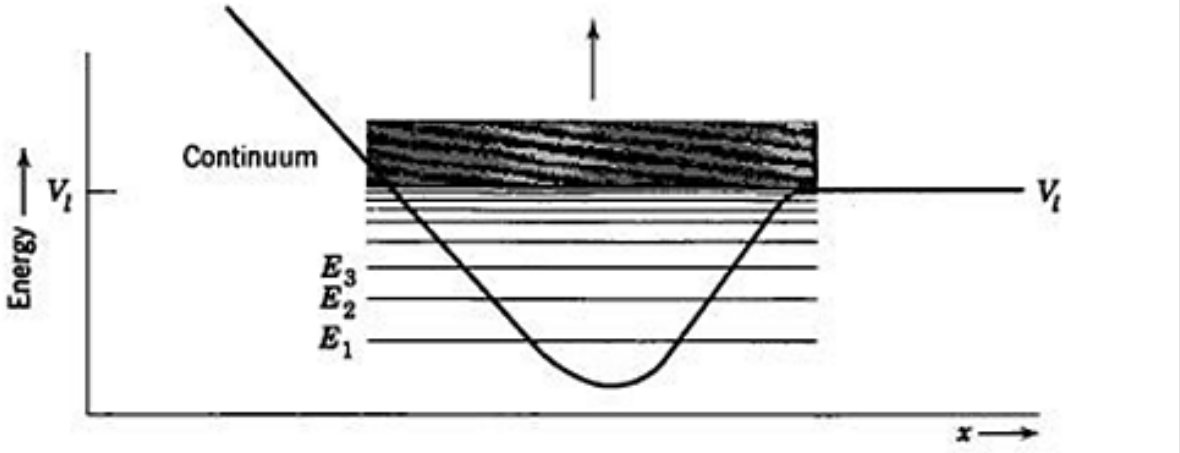


Figure 7–5. The discrete and continuum values of allowed energy for a typical time independent Schroedinger equation.

When that relation is such that the particle is not bound to a limited region, then the theory predicts the total energy can have any value.